

AI for Redistricting — Spring 2026

Gerrymandering Analysis of Massachusetts Congressional Districts

2024 Presidential & Senate Elections | ReCom Ensemble Analysis

Massachusetts Population Context

9

Congressional Districts

7.03M

Total Population (2020)

100%

Dem-held seats (current)

D+25

2024 Pres partisan lean

Partisan Landscape

Democrats hold all 9 congressional seats. Harris won Massachusetts by about 25 points in 2024 (61.2% to 36.0%), so the baseline electorate is already strongly Democratic.

Racial Landscape

Massachusetts is majority white (~71%), with minority populations concentrated in Boston and nearby cities, anchoring District 7 as a majority-minority district.

District Geography

The districts cover very different regions: D1 western MA, D2 central/west, D6 North Shore, D7 Boston/Cambridge/Somerville, D8 Boston/South Shore, and D9 South Coast/Cape/Islands.

Population Pattern

Because Democratic voters are geographically dense around Boston and other urban areas, many random plans still elect Democrats; the key question is how much the actual map packs those voters.

Massachusetts Legal Context

2021

Current map enacted

State Leg.

draws districts

No IRC

no independent commission

Gov. veto

ordinary statute

Map Creation

The current congressional map was enacted in 2021 by the Democratic-controlled Massachusetts Legislature, with no independent redistricting commission.

Redistricting Process

Massachusetts uses the normal legislative process: the General Court draws maps, the governor can veto, and vetoes can be overridden by a two-thirds vote.

Legal Landscape

I did not find a major lawsuit against the 2021 congressional map itself. The concern is more political: all 9 seats are Democratic in a strongly Democratic state.

Notable Litigation

Related Massachusetts cases include Boston's 2023 City Council map being blocked over race-based districting concerns, and earlier Boston-area state House litigation over minority vote dilution.

Data & Methods

Data Sources

1

2020 Census Blocks

107,278 census blocks with total population (P0010001). Used to aggregate population to precinct level.

2

MA County Shapefile

14 counties. Used to nest precincts within counties during MAUP repair.

3

2024 General Election Precincts

2,383 precincts with vote totals for Presidential (Harris/Trump) and Senate races.

MAUP Data Cleaning Pipeline

1

Reproject

Aligned all three shapefiles for accurate spatial operations.

2

smart_repair

Fixed 519 overlaps, 77 holes, and invalid geometries in election precincts, nesting within county boundaries.

3

Rook → Queen

Converted adjacencies < 30m to point adjacencies so GerryChain treats them as proper neighbors.

4

Assign Population

Used `maup.assign()` to map 107k census blocks → 2,383 precincts and sum population.

5

Validation

`maup.doctor()` returns True. Block total = Precinct total = 7,029,917.

Ensemble Method: ReCom

How ReCom Works

- 1 Pick two adjacent districts at random.
- 2 Merge them into one region.
- 3 Build a random spanning tree of the merged region.
- 4 Cut one edge of the spanning tree to create two new valid districts satisfying population constraints.
- 5 Repeat 2,000 times to generate the ensemble.

Parameter Choices

Steps per chain	20000
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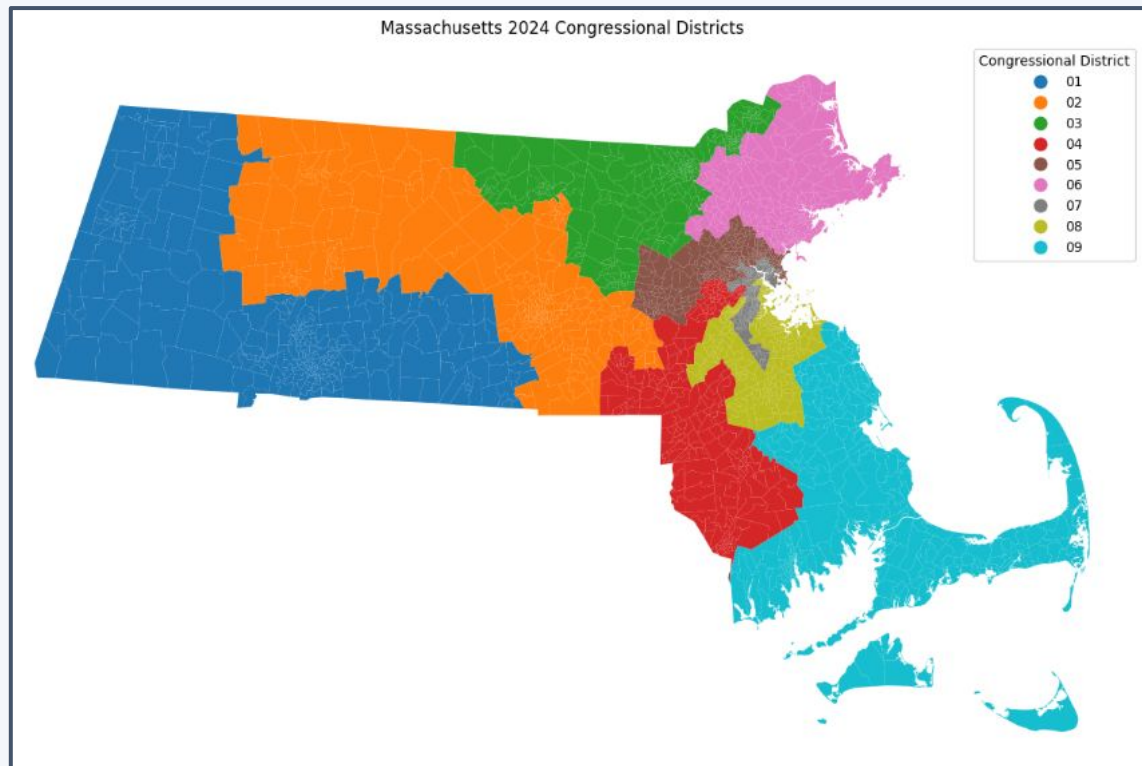
Number of chains	2 (Pres + Senate)
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Population tolerance	$\pm 2\%$
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Node repeats	2
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Compactness bound	$\leq 2 \times$ initial cut edges
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The Actual Massachusetts Map



9 districts

All Democrat-held

659

Cut edges in actual map

-0.24

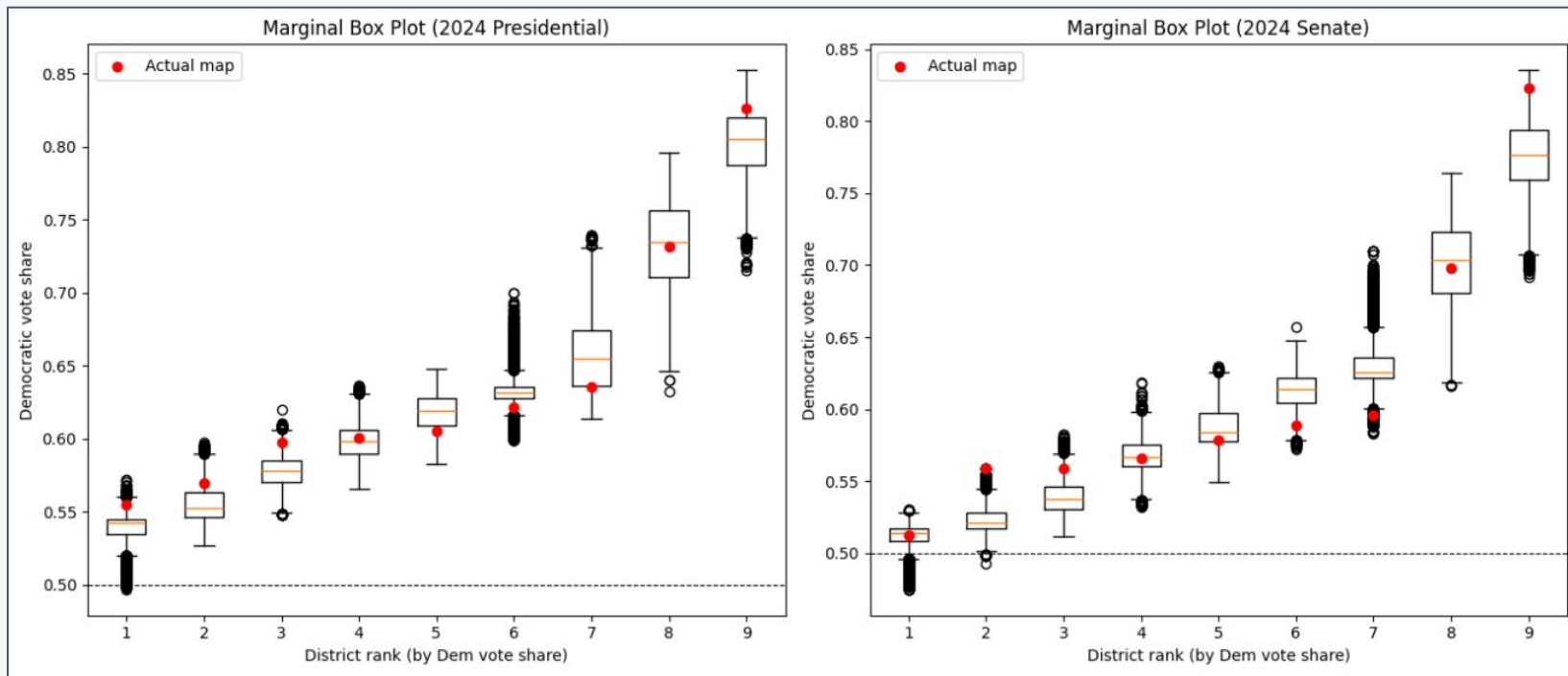
Efficiency Gap (Pres)

-0.30

Efficiency Gap (Senate)

Outlier Analysis: Marginal Box Plots

Red dots = actual MA map. Each box shows the simulated range for the 1st through 9th most Democratic district, after sorting each plan by Democratic vote share.



Key finding: The actual map is ordinary in the lower/middle ranks, but high at the Democratic end. The 9th box is the most Democratic district by vote share, not Congressional District 9; from the map, that likely reflects the Boston/Cambridge/Somerville core (D7), while D9 is the South Coast/Cape/Islands seat.

Understanding the Efficiency Gap

What is the Efficiency Gap?

Measures wasted votes — all losing-party votes are wasted, plus any winning-party votes beyond the 50%+1 threshold needed to win.

$$\text{EG} = (\text{Wasted Dem} - \text{Wasted Rep}) / \text{Total Votes}$$

Negative EG → map favors Democrats

Positive EG → map favors Republicans

EG Results

Election	Ensemble Mean	Actual Map
Pres 2024	-0.12	-0.24
Senate 2024	-0.14	-0.30

Compactness: Polsby-Popper Score

Polsby-Popper

$$PP = 4\pi \times \text{Area} / \text{Perimeter}^2$$

Score of 1.0 = perfect circle (most compact)

Score near 0 = very irregular or fragmented shape

The actual MA map has more cut edges (659) vs ensemble mean (~480), suggesting deliberately fragmented boundaries.

Actual Map PP Scores

D01  0.2797

D02  0.2119

D03  0.3264

D04  0.2341

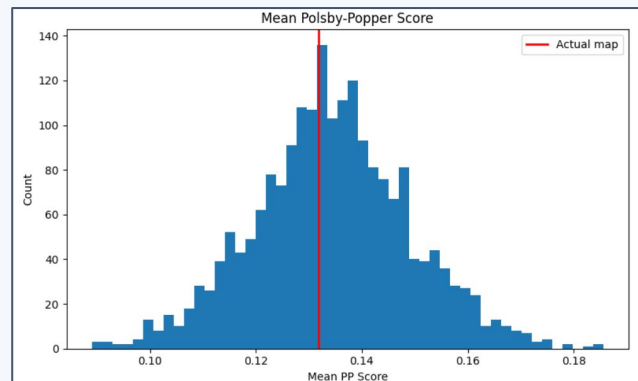
D05  0.3876

D06  0.2654

D07  0.1823

D08  0.1934

D09  0.2512



Conclusions

Extreme Efficiency Gap

The actual MA map shows an EG of -24% (Pres) and -30% (Senate), both in the extreme left tail of the ensemble — far more Democratic-favoring than nearly all 2,000 randomly generated maps.

All 9 Districts Democratic

Winning all 9 districts is a rare outcome in the ensemble. Some of this comes from the state's Democratic lean, but the actual map still performs unusually strongly for Democrats.

More Fragmented Borders

659 cut edges vs ensemble mean of ~ 480 indicates the map has more fragmented district boundaries than typical — consistent with deliberate line-drawing.

Rank-9 / Urban Packing

The box plots sort districts by vote share. “Rank 9” means the bluest district, not CD9. The high red dot likely reflects Boston/Cambridge/Somerville-area voters being packed into the bluest seat.